METRIC

MIL-N-62418(AT)
6 December 1992
SUPERSEDING
(see 6.6)

MILITARY SPECIFICATION

NICKEL BASE ALLOY, SHEET AND PLATE CORROSION AND HEAT RESISTANT, SOLUTION HEAT TREATED (HAYNES 230) (METRIC)

This specification is approved for use by the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the chemical, mechanical property, and inspection requirements for sheet and plate of this alloy.

1.2 Classification. The nickel base alloy covered in this specification are classified by type as follows (see 3.1 and 6.2).

Type Ia - Hot rolled sheet
Type Ib - Cold rolled sheet
Type II - Hot rolled plate

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 9630

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STANDARDS

FEDERAL

FED-STD-151	- Metals; Test Methods.
FED-STD-183	- Continuous Identification Marking of Iron and Steel Products.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Printing Service Office, Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the document cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8	- Standard Test Methods for Tension Testing of Metallic Materials.
ASTM E112	- Standard Test Methods for Determining Average Grain Size.
ASTM E139	- Standard Practices for Conducting Creep, Creep-Rupture, and Stress Rupture Tests of Metallic Materials.
ASTM E290	- Standard Test Method for Semi Guided Bend Test for Ductility of Metallic Materials.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 2262	- Tolerances Nickel, Nickel Alloy, and Cobalt Alloy Sheet, Strip, and Plate.
AMS 2269	- Chemical Check Analysis Limits Wrought Nickel Alloys and Cobalt Alloys.
AMS 2371	- Quality Assurance Sampling of Corrosion and Heat Resistant Steels and Alloys Wrought Products Except Forgings and Forging Stock.

(Applications for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The nickel base alloy shall come in three types (see 1.2 and 6.2) of material and shall be solution heat treated. Solution heat treatment shall be accomplished by heating the material at 1175-1245°C (2150-2275°F) and holding at this temperature for a time commensurate with the section thickness of the material. The annealing period will last as long as necessary to produce material that conforms to this specification.

3.1.1 Finish. The surface of the sheet or plate shall be delivered descaled unless solutionizing produces a bright finish of equal or better quality than the no. 1 finish for corrosion resistant sheet.

3.1.2 Chemical composition. Chemical composition of the material shall conform to the percentage by weight for each element specified in table I.

TABLE I. Chemical composition (Haynes 230).

Elements	Percentage by weight	
	Minimum	Maximum
Carbon	0.05	0.15
Manganese	0.30	1.00
Silicon	0.25	0.75
Phosphorus	--	0.015
Sulfur	--	0.015
Chromium	20.00	24.00
Cobalt 1/	--	5.00
Tungsten	13.00	15.00
Molybdenum	1.00	3.00
Aluminum	0.20	0.50
Iron	--	3.00
Lanthanum	0.005	0.05
Boron	--	0.015
Nickel	Remainder	

1/ Cobalt is the sum percentage of cobalt present in the alloying elements used to form this material, and is not additive to the formulation.

3.1.2.1 Chemical composition variation. Allowable variation of the chemical composition shall be in accordance with the requirements of AMS 2269.

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3.1.3 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.3.1 and 6.3.2).

3.2 Tolerances. Unless otherwise specified in the contract or purchase order (see 6.2) tolerances shall be in accordance with the requirements of AMS 2262.

3.3 Performance. The material shall conform to the following mechanical and physical properties after heat treatment.

3.3.1 Mechanical properties. Mechanical properties of the heat treated material shall be as specified herein.

3.3.1.1 Tensile. The room temperature tensile properties of the nickel base alloy shall be as follows (see 4.3.5):

	Megapascals	(PSI)
Ultimate strength, minimum:	758	110,000
Yield strength at 0.2% offset, minimum:	310	45,000

Elongation in 50.8 mm (2 inch), minimum thickness .038 mm (.015 inch) and over	Elongation	
	40%	40%

NOTE: For material thicknesses less then .38 mm (0.015 inch), elongation shall be as agreed upon by the procuring activity and the supplier.

3.3.1.2 Stress rupture. A test specimen maintained at a temperature of 927 ± 2°C (1700 ± 3°F), while the axial load specified below is applied continuously, shall not rupture in less than 24 hours. Elongation after rupture, measured at room temperature, shall be 10 percent for thicknesses over .038 mm (.015 inch).

NOTE: For thickness less than .38 mm (.015 inch), elongation shall be as agreed upon by the procuring activity and the supplier.

3.3.1.3 Bending. All bending shall be done at room temperature by aligning rolling direction parallel to the bending axis, and wrapping the material through an angle of 3.15 radians (180 degrees). To calculate bend radius multiply the nominal material thickness by the appropriate bend factor listed below. No cracking is permitted.

Nominal thickness	Bend factor
Up to 1.25 mm (.050 inch)	1.5
1.25 to 4.75 mm (.050 to 0.187 inch)	2.0
4.75 mm and above	As specified in contract or purchase order

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3.3.2 Physical properties. Physical properties of the heat treated material shall be as specified herein.

3.3.2.1 Grain size. The grain size of the nickel base alloy shall be as follows (see 4.3.8):

Nominal thickness mm (inch)	ASTM Grain size
Up to 3.2 mm (.125 inch)	4 or finer
Over 3.2 mm (.125 inch)	3 or finer

3.3.2.2 Oxidation resistance. A polished cross-section of the test specimen shall have not more than .04 mm (.0015 inch) of metal converted to oxide scale, plus any continuous intergranular oxidation, when prepared and tested in accordance with 4.3.9.

3.4 Identification marking. Each sheet or plate shall be identified in accordance with FED-STD-183 (see 6.2). The marking shall include this specification number, revision letter, heat number, manufacture's identification, and nominal thickness. When the identification marking is performed directly on the material, the marking fluid shall not have any deleterious effect on the material. The marking fluid shall be capable of being removed in a hot alkaline solution (see 4.3.2).

3.5 Workmanship. The nickel base alloy shall be uniform in quality and condition, clean, sound, and free from foreign materials, and from internal and external imperfections detrimental to the fabrication or performance of the parts (see 4.3.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Quality conformance inspection (QCI). QCI shall include the examination of 4.2.1 and the tests of 4.3.3. Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection of the sample and the inspection lot (see 6.3).

4.2.1 Examination. All material shall be inspected to assure compliance with requirements specified in 3.2, 3.4 and 3.5.

4.2.2 Lot formation and sample for tests. Unless otherwise specified in the contract or purchase order (see 6.2) lot formation and sample for tests shall be in accordance with AMS 2371. The sample shall be subjected to the tests specified in 4.3.4 thru 4.3.9. The oxidation resistance test (see 4.3.9) shall be performed on the first lot of material from each heat.

4.2.2.1 Tensile test sample. Tensile test sample taken from widths 230 mm (9 inches) and over shall be taken with the axis perpendicular to the direction of rolling. For widths less than 230 mm (9 inches), tensile test sample shall be taken with the axis parallel to the direction of rolling.

4.2.2.2 QCI failure. Any sample that fails to conform to any specified requirement shall be rejected: any failure (one or more) of the selected sample for the appropriate inspection lot size shall constitute a failure of the entire lot.

4.3 Methods of inspection.

4.3.1 Materials. Conformance to 3.1.1 and 3.1.3 shall be determined by inspection of contractor records providing proof or certification that processing and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certification, industry standard, test reports, and rating data.

4.3.2 Defect. Conformance to 3.4 and 3.5 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

TABLE II. Classification of defects.

Category	Characteristics	Method of inspection
Critical	None	
101	Marking incomplete/improper (see 3.4).	Visual
102	Workmanship faulty (3.5).	Visual

4.3.3 Test. All test shall be conducted on the material after solution heat treatment of the material.

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4.3.4 Chemical composition test. The sample shall be analyzed to verify conformance to 3.1.2. The chemical composition analysis shall be in accordance with Method 111 or 112 of FED-STD-151.

4.3.5 Tensile test. Test the sample in accordance with ASTM E8 to verify conformance with 3.3.1.1.

4.3.6 Stress rupture test. Test the sample in accordance with ASTM E139 to verify conformance with 3.3.1.2.

4.3.7 Bending test. Test the sample in accordance with ASTM E290 to verify conformance with 3.3.1.3.

4.3.8 Grain size test. Samples shall be sectioned and polished to appropriate fineness by metallographic methods and suitably etched to reveal grain structure. The grain size shall be determined in accordance with ASTM E112. The grain size shall meet the requirements of 3.3.2.1.

4.3.9 Oxidation resistance test. The sample shall be prepared and tested as follows to determine conformance with 3.3.2.2.

a. The sample shall have at least 1000 mm^2 (1.5 square inches) in excess of the test surface required for fixturing. The test surface shall be hand polished, using a silicon carbide grit of 120 or finer, and suitably cleaned.

b. The sample shall be subjected to four thermal cycles consisting of heating in still air for 25 $\pm$ 1 hours, at a selected temperature between 1095-1150 $\pm$ 15°C (2000-2100 $\pm$ 25°F) followed by cooling to 150°C (300°F) or lower between cycles.

c. Samples with localized areas of bare metal (spalled scale) greater than 1.5 mm (0.062 inches) in diameter shall be considered invalid, unless the condition can be attributed to contact with ceramic supports. Invalid material shall be retested. If the condition persists on the retest, the material shall be subject to rejection.

4.3.9.1 Spalling. To determine if spalling is in excess of 1.5 mm the test sample shall be measured before and after the four thermocycles of heating. Care shall be taken not to remove the scale before measuring as it is not very adherent and can be easily removed.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This material is used primarily for formed and drawn parts requiring high strength up to 980°C (1800°F), and oxidation resistance up to 1095°C (2000°F).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of nickel base alloy plate or sheet.
c. Issue of the DODISS to be cited in the solicitation and if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. If tolerances shall be other than as specified (see 3.2).
e. Bend factor for material greater than 4.75 mm (see 3.3.1.3).
f. Identification marking (see 3.4).
g. If responsibility for inspection is other than as specified (see 4.1).
h. If lot formation and sample for tests shall be other than specified (see 4.2.2).
i. Selection of applicable level and packaging requirements (see 5).

6.3 Definitions.

6.3.1 Definitions of terms used in sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. Defective. A defective is a unit of product which contains one or more defects.

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e. Formation of lots or batches. The product shall be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 1). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "inspection lot" or "inspection batch", i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects; and may also contain minor defects but contains no critical defect.

j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

n. Sample A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

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o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.3.2 Recovered material. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.3).

6.3.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Material Command (AMC) policy for the elimination of the AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) for military specifications.

6.5 Subject term (key word) listing.

Bending	Oxidation
Chemical composition	Solution heat treatment
Elongation	Stress rupture
Material	Tensile

6.6 Supersession data. This military specification supersedes Textron Lycoming internal specification M3633B, dated 13 June 1991.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 9630-A024)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-N-62418(AT)	921206

3. DOCUMENT TITLE Nickel Base Alloy, Sheet and Plate Corrosion and heat Resistant, Solution Heat Treated (Haynes 230) (Metric)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(313) 574-5954	DSN 786-5954
c. ADDRESS (Include Zip Code) COMMANDER U.S. ARMY TANK - AUTOMOTIVE COMMAND ATTN: AMSTA-GDS WARREN, MICHIGAN 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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